

# Theoretical Computer Science, exam of 2025.01.13

SURNAME \_\_\_\_\_ NAME \_\_\_\_\_ PERSONAL CODE \_\_\_\_\_

Total available time: 1h 30'. You can use books and paper notes during the exam, while electronic devices are prohibited. Each correct exercise is valued 11 points. Write the results in this paper.

Multichance students do not need to answer to Exercises 2.2 and 3.2: bar here in that case [ ]

## Exercise 1

Consider the language  $L = \{x.y \mid x, y \in \{a, b\}^+, |x| = 2|y| \text{ or } 2|x| = |y|\}$ .

Define a formal model for L with minimum power.

## Exercise 2

1) Consider the following function: 
$$g(y) = \begin{cases} 2y & \text{if } f_y(y) = y \\ f_y(y) & \text{otherwise.} \end{cases}$$

Tell if g is a total function and if it is computable, clearly stating why.

2) Let A be a semi-decidable set, and B a decidable set. It is easy to see that  $C = A - B$  is semi-decidable. For this reason, define a computable (possibly partial) function having C as its image.

### Exercise 3

Consider the language  $L = \{x.h(x) \mid x \in \{a, b\}^+\}$ , where the function  $h$  is the homomorphism  $h(a) = b$  and  $h(b) = a$ , e.g.  $h(abaa) = babb$ .

1) Describe a *single tape Turing machine* and 2) a *RAM machine* for  $L$ , evaluating their spatial and temporal complexities.

# Solutions

## Exercise 1

$L$  is a regular language, since it can be written as  $L = \{x \in \{a, b\}^+ \mid |x| \bmod 3 = 0\}$ .

The following regular grammar generates  $L$ :

$S \rightarrow aA \mid bA$

$A \rightarrow aB \mid bB$

$B \rightarrow a \mid b \mid aS \mid bS$

## Exercise 2

1) First of all notice that there always exist a  $y$  such that  $f_y(y)$  is undefined (take for instance the  $y$  for the totally undefined function). Hence we can say that  $g$  is *partial and computable*: we just simulate MT  $y$  with  $y$  as input. If this computation terminates, we multiply the result by 2; if it doesn't terminate, that's fine by definition.

2)  $g_C(x) = \text{if } C_B(g_A(x)) = 0 \text{ then } g_A(x) \text{ else } \perp$

## Exercise 2

The single-tape Turing machine moves back and forth, marking the symbols encountered from the ends toward the center (rejecting the input if the length of the string is not even). At this point, it rewinds the tape, removing the marks from the first half of the symbols. Finally, it checks the correspondence between the symbols in the two halves by moving back and forth, marking the symbols in the first half again and replacing those in the second half with a blank. The time complexity is  $\Theta(n^2)$ , and the space complexity is  $\Theta(n)$ , where  $n$  is the length of the input string.

The RAM machine reads the string into memory, using a counter to calculate its length,  $n$ . It then verifies the correspondence between the two halves using two counters: one starting at 0 and the other at  $n/2$ , using direct access. With the constant cost criterion, the space complexity is  $O(n)$  (2 cells for the counters and  $O(n)$  cells for the string), and the time complexity is  $O(n)$ . With the logarithmic cost criterion, the space complexity remains  $O(n)$  (even though the counters occupy  $O(\log(n))$  cells), while the time complexity is  $O(n \log(n))$  due to the counter.